Data Cleaning,Analysis&Visualization with Excel

Project Objective:

The aim of the project is to perform Data cleaning and use of excel formulas and pivot tables for further analysis with data visualization for clear insights for this dataset.

Dataset Description:

The dataset has four tables which are Customer Dimension,store dimension,product dimension and sales fact table.

Column Description:

Column Name	Description
Quarter	To find Quarter from order date
Month	To find month from order date
year	To find year from order date
Stock range	To classify stock according to their values

Data cleaning:

Duplicate values are removed using remove duplicates.In customer dimension table empty columns in loyalty level is replaced with unknown for text values.Inappropriate texts are arranged using Proper option.Extra spaces are removed using Trim and clean options. Missing values in Total amount and unit price columns are calculated by taking average of the columns.

Order Date column is split into three columns as Quarter,Month,Year for deriving insights.

Highlighting cells and conditional formatting:In customer dimension table age column is highlighted for above 30 years of age.
Color bars is applied in cost column to define their price ranges.

If function:

IF function is applied on stock value column to define the stock ranges.
Formula used:IF(G2>200,”High”,”Low”)

SumIf function:

SUMIF function is used to sum the values in accessories sub category.
Formula Used:SUMIF(C2:C101,C7,E2:E101)=3708.07

Sum:

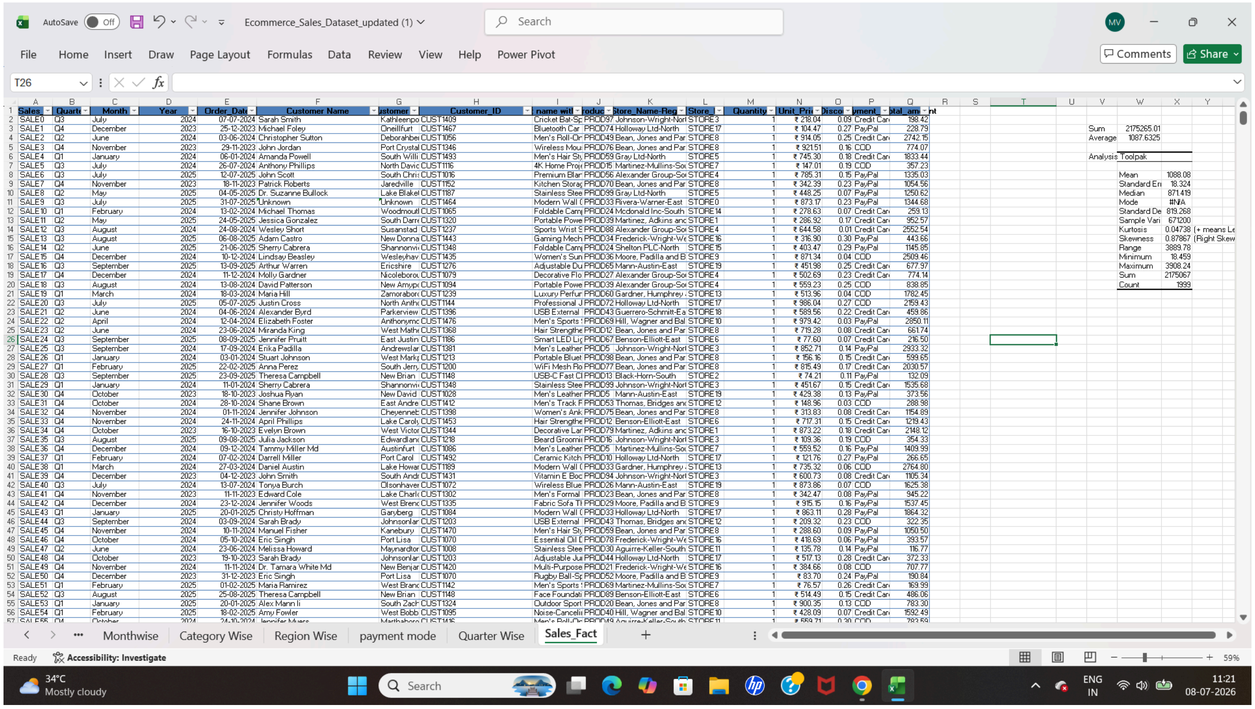
Sum is calculated for total amount in sales fact table
Formula used:SUM(Q2:Q2001)=2175265.01

Average:

Average is calculated for total amount in sales fact table
Formula used:AVERAGE=SUM/2000=1087.6325

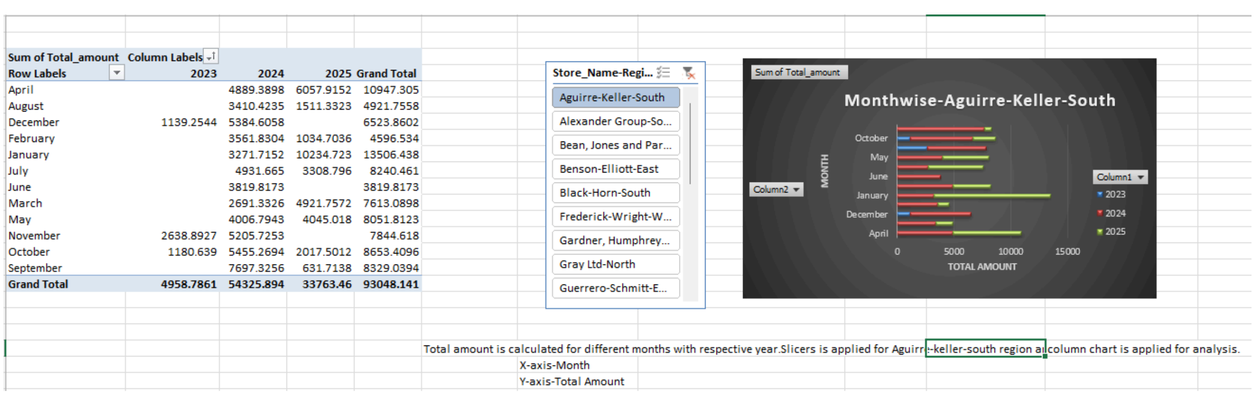
Analysis Toolpak:

Analysis toolpak is used for calculating mean, median.mode and some statistical values.
Mean:1088.077
Standard Deviation:819.2682
Median:871.4193
Mode:no value
Standard error:18.32398
Sample Variance:671200.4
Kurtosis:0.047375
The output is positive so it is Leptokurtic.
Skewness:0.878674
Skewness is also positive value(right skew)
Mean>Median>Mode
Range:3889.784
Minimum:18.459
Maximum:3908.243
Count:1999



Pivot Table:

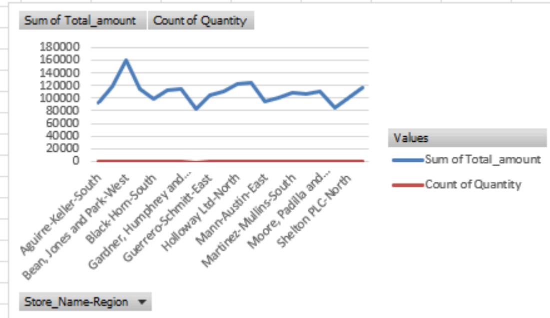
Pivot table is used for summarize,analyse this dataset and charts are created for further analysis.



Sum of Total_amount	Column Labels			
Row Labels	2023	2024	2025	Grand Total
April		4889.3898	6057.9152	10947.305
August		3410.4235	1511.3323	4921.7558
December	1139.2544	5384.6058		6523.8602
February		3561.8304	1034.7036	4596.534
January		3271.7152	10234.723	13506.438
July		4931.665	3308.796	8240.461
June		3819.8173		3819.8173
March		2691.3326	4921.7572	7613.0898
May		4006.7943	4045.018	8051.8123
November	2638.8927	5205.7253		7844.618
October	1180.639	5455.2694	2017.5012	8653.4096
September		7697.3256	631.7138	8329.0394
Grand Total	4958.7861	54325.894	33763.46	93048.141

This is pivot table for monthly analysis of the sales.In this bar chart Slicers is also applied to display sales in specific region.

w Labels	Sum of Total_amount	Count of Quantity
uirre-Keller-South	93048.1405	88
exander Group-South	118814.1157	102
an, Jones and Park-West	158971.4624	127
nson-Elliott-East	114533.7966	106
ack-Horn-South	98960.8078	92
ederick-Wright-West	111704.9436	106
ardner, Humphrey and Contreras-Ea	114401.4715	95
ay Ltd-North	82232.34831	83
uerrero-Schmitt-East	104313.0545	105
ll, Wagner and Ballard-East	109454.5741	97
olloway Ltd-North	122531.7114	110
hnson-Wright-North	125011.8285	114
ann-Austin-East	94189.25091	94
artinez, Adkins and Johnson-North	101060.6092	94
artinez-Mullins-South	108039.3494	108
cdonald Inc-South	106248.2833	95
oore, Padilla and Butler-South	110102.399	96
vera-Warner-East	85138.5552	93
elton PLC-North	99679.086	95
omas, Bridges and Harris-North	116829.2206	100
and Total	2175265.009	2000

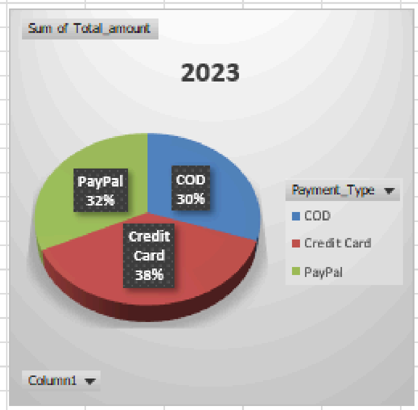


Regionwise sales for different regions is calculated using Line chart
X-axis-Sum of total amount,count of quantity
Y-axis-Store name-region

This pivot table is for region wise total sales with Line chart.

Sum of Total_am	Column La			
Row Labels	2023	2024	2025	Grand Total
COD	67845.31492	358305.665	277552.36	703703.339
Credit Card	84467.0312	319429.08	310947.458	714843.57
PayPal	71855.5427	416140.578	268721.979	756718.099
Grand Total	224167.8888	1093875	857221.8	2175265

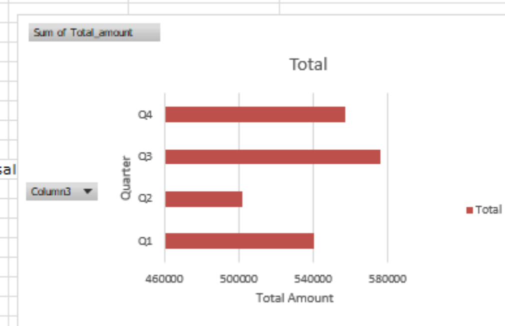
Pie chart is created for payment type with respective sum of total amount
From this chart,most of payment done with credit card type.



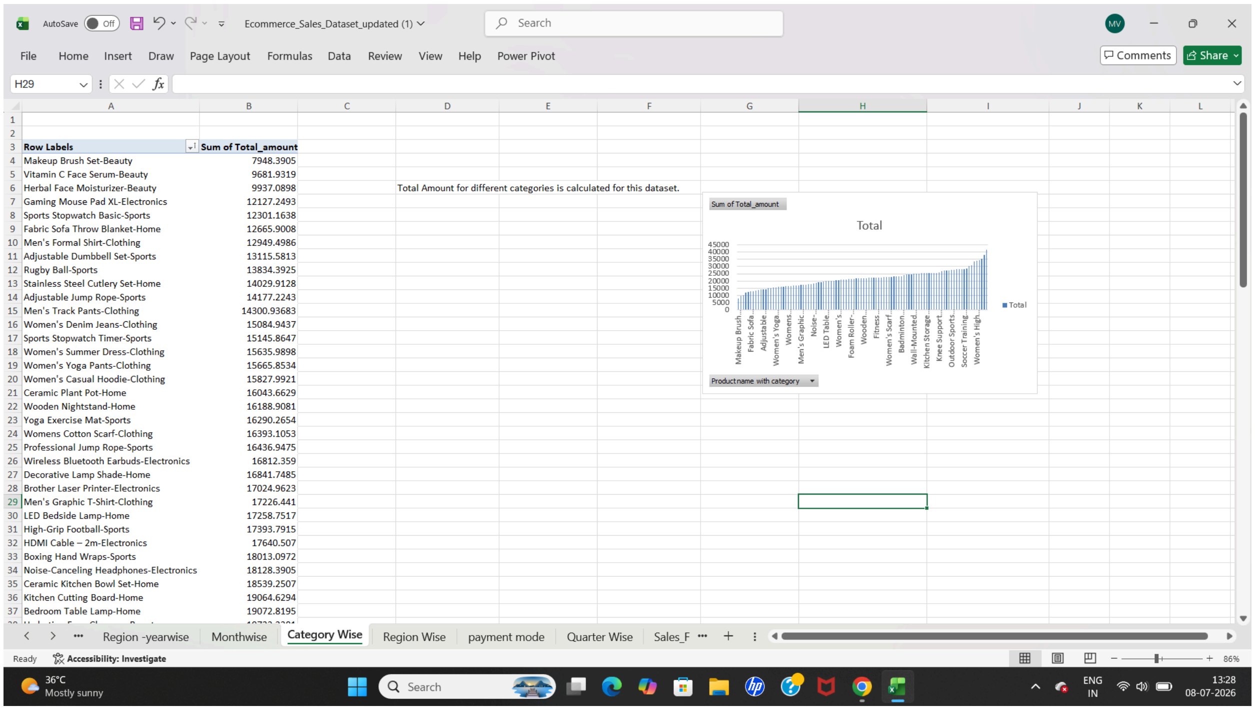
This table denotes type of payment with respective total amount by use of pie chart.

ow Labels	Sum of Total_amount
Q1	540227.6011
Q2	502016.0147
Q3	575956.871
Q4	557064.5218
rand Total	2175265.009

um of total amount is calculated for different quarters.From this Quarter-4 has higher amount of sal

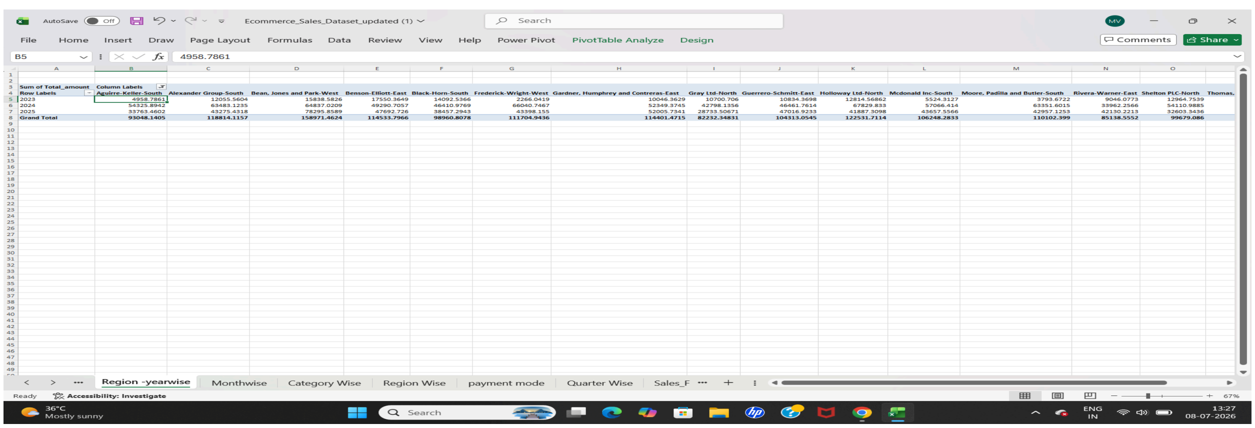


This table has total amount of sales of different quarters. With help of column chart.



Row Labels	Sum of Total_amount
Makeup Brush Set-Beauty	7948.3905
Vitamin C Face Serum-Beauty	9681.9319
Herbal Face Moisturizer-Beauty	9937.0898
Gaming Mouse Pad XL-Electronics	12127.2493
Sports Stopwatch Basic-Sports	12301.1638
Fabric Sofa Throw Blanket-Home	12665.9008
Men's Formal Shirt-Clothing	12949.4986
Adjustable Dumbbell Set-Sports	13115.5813
Rugby Ball-Sports	13834.3925
Stainless Steel Cutlery Set-Home	14029.9128
Adjustable Jump Rope-Sports	14177.2243
Men's Track Pants-Clothing	14300.93683
Women's Denim Jeans-Clothing	15084.9437
Sports Stopwatch Timer-Sports	15145.8647
Women's Summer Dress-Clothing	15635.9898
Women's Yoga Pants-Clothing	15665.8534
Women's Casual Hoodie-Clothing	15827.9921
Ceramic Plant Pot-Home	16043.6629
Wooden Nightstand-Home	16188.9081
Yoga Exercise Mat-Sports	16290.2654
Womens Cotton Scarf-Clothing	16393.1053
Professional Jump Rope-Sports	16436.9475
Wireless Bluetooth Earbuds-Electronics	16812.359
Decorative Lamp Shade-Home	16841.7485
Brother Laser Printer-Electronics	17024.9623
Men's Graphic T-Shirt-Clothing	17226.441
LED Bedside Lamp-Home	17258.7517
High-Grip Football-Sports	17393.7915
HDMI Cable – 2m-Electronics	17640.507
Boxing Hand Wraps-Sports	18013.0972
Noise-Canceling Headphones-Electronics	18128.3905
Ceramic Kitchen Bowl Set-Home	18539.2507
Kitchen Cutting Board-Home	19064.6294
Bedroom Table Lamp-Home	19072.8195

Sales for different products and their sub categories.



Yearwise sales for different regions.

<u>Descriptive Analytics:</u>

It describes what happened in the dataset.

1.Which month generates highest revenue:January Month

2.Which region contributes highest revenue:Bean,Jones and Park-west.

3.Which payment method is most popular:Credit Card.

4.Which year sales is higher:2024

Diagnostic Analytics:
It describes why did it happen.
1.Which city have lowest sales:Shelton PLC
2.Which quarters have lowest performance:Quarter-2
3.Which sub category has highest amount of sales:Electronics
4.Which region has lowest quantity:North

Predictive Analytics:
What is going to happen in future.
1.Which product may have increased demand:Electronics product.
2.Which year sales to be focused:2023
3.Which month have lowest sales:June

Prescriptive Analytics:
What should we do for enhancement.
1.Which type of stores need improvement:Outlet
2.Which payment type should be promoted:COD
3.Which quarters need to be increased:Q-2

Conclusion:

Data is cleaned,analysed and summarized with help of excel functions.Visualization is also done with pivot tables and charts.